

Varietal Seed Classification and Seed Germination Prediction System

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Abstract— Seed classification and germination prediction are crucial aspects of modern agriculture and ecosystem restoration. Seeds are vital for agriculture and biodiversity conservation, and their accurate classification and germination prediction are crucial for sustainable farming practices. This literature survey paper presents a detailed summary of various machine learning techniques employed for performing seed classification and germination prediction. The paper reviews the latest research, methodologies, and datasets employed in performing seed classification and germination prediction tasks. It also highlights the challenges and opportunities in this domain, including the integration of advanced deep learning models, multispectral imaging, and sensor technologies. The objective of our research is to investigate the impact of environmental factors on the germination potential of diverse seed types, classify seeds based on their physiological characteristics, and develop predictive models for estimating germination timing under varying conditions. By synthesizing existing knowledge, this review aims to establish a basis for future research and advancements in seed-related applications of machine learning.

Keywords— seed classification, germination prediction, machine learning, deep learning, multispectral imaging

I. INTRODUCTION

The use of computers to predict the type of grain is a relatively new field that has made significant progress recently. This review aims to look at how grains are classified and what problems still need solving. Predicting the type of grain is important for farmers because it helps them to decide what crops to plant and how to take care of them in an efficient manner. In recent years, many researchers have worked on this topic. In this review, we will summarize what researchers have found so far and talk about the problems that still need to be addressed. As the world's population grows, it's more important than ever to produce food efficiently. In this review, we will explore how computers can help predict the type of grain. We will start by talking about the different computer methods and how they are used. Then, we will discuss the different things researchers look at while predicting the grain type. Finally, we will compare how well different computer

methodologies work for predicting the grain type. The development of a pip is a very important progression among the various growing phases of shrubs and vegetation. It is a complex biological phenomenon influenced by a multitude of factors, including environmental conditions, seed genetics, and physiological traits. This paper presents a broad exploration of seed germination prediction, delving into the existing research, methodologies, and challenges. Moreover, we present a novel approach to the germination prediction of seeds by leveraging various advanced machine-learning techniques and large-scale datasets. Our proposed model considers a wide range of factors, including the characteristics of seeds, environmental conditions, and also the historical germination data. By integrating these crucial factors, we strive to create a robust and accurate prediction tool that can be applied across diverse species of plants and ecosystems.

II. LITERATURE REVIEW

In the reference [1], the authors wanted to curb the illegal trade practices of rice grains. They explored the various reasons and also used numerous technicalities. The authors used numerous image processing techniques and algorithms like LiClipse, Eclipse, OpenCV, and Spider. One of the primary objectives of this research is to assist the rice vendors in preventing fraudulent transportation and sales of rice.

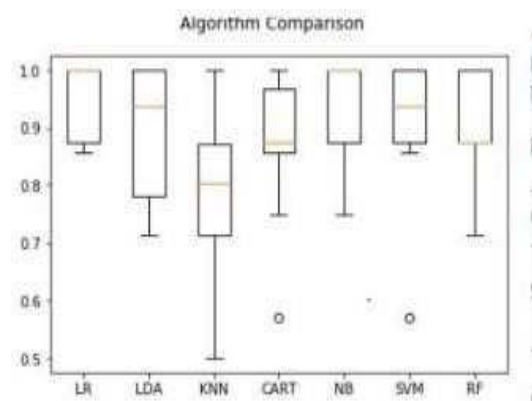


Fig. 1. Box plot of various algorithms

Figure 1. displays the box plot of several algorithms.

In reference [2], the authors introduce a novel deep-learning method for identifying the various grain types. The approach proposed by the authors integrates the strengths of two of the distinct and different deep learning models namely Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN). A convolutional Neural Network is employed to extract the generic features from the various grain types. A recurrent Neural Network is employed to study the particular features that tell apart various grain types. The authors estimated their indigenous approach on a curated dataset of thirteen different grain types and achieved an overall accuracy of 96.12 %. The authors' method portrayed greater performance than the established and top-notch techniques like the K-Nearest Neighbour (KNN) classifier and other traditional Convolutional Neural Networks.

In the reference [3], the authors review the recent developments and techniques in rice grain analysis, classification, and quality assessment with an emphasis on the novel methods used. This helps to assure fraudulent free transport of grains to international markets. The authors also delve into the various aspects of this proprietary technology which can assist to assure genuine classification and rating of grains. The authors employed the Support Vector Machine technique on a curated dataset of images and achieved a very high precision. The study also incorporated various other techniques such as Fourier transform and DCNN to further improve its capabilities.

In the reference [4], the authors proposed a new method for identifying and diagnosing pests and diseases on sesame seeds using image classification. The authors employed a well-known classification technique known as Maximum Margin Classifier to classify with remarkable accuracy and efficiency. By using this technique, the authors were able to gauge the robustness of the classifier. The authors compared their custom-developed classifier to five pre-existing models, including variants of VGG and ResNet. The classifier exhibited remarkable results boasting a very high and impressive precision.

In the reference [5], the authors projected an innovative technique for performing segregation with the help of agglomerative grouping and a dynamic deep learning model. The authors applied this technique for segregation based on shape and irregular features. The authors first used agglomerative grouping to group similar kernels together. Then, they used a dynamic deep-learning model to identify the specific grain types within each group. The authors evaluated their method on a dataset of corn kernels and achieved an average accuracy of over 90 %, demonstrating the effectiveness of their approach.

In the reference [6], the authors propose a new method for automated seed recognition and categorization. Their method uses a combination of image processing, feature extraction, and classification. They evaluated their method on a dataset of 400 seeds from four different species: oats, corn, lentils, and barley. The authors first perform a series of image pre-processing steps, including dimension reduction using PCA and clustering using K-means. Then, they extract features from the seeds using shape and texture analysis. Finally, they classify the seeds using the nearest Euclidean distance rule. The authors achieved an identification rate of 85.75 % using shape features, 78 % using texture features, and 89.25 %

using a combination of shape and texture features. This demonstrates that their method can effectively recognize and categorize seeds.

In the reference [7], the authors propose a new method for classifying cotton seed grades based on the least squares method and support vector machines (SVMs). They first use image processing algorithms to extract features from images of cotton seeds, such as impurity levels, yellowness, and brightness. Then, they use the least squares method and SVMs to classify the cotton seed into different grades. The authors evaluate their method on a dataset of cotton seeds and achieve an accuracy of 92 %, which is significantly higher than the accuracy of 83 % achieved by existing models. This demonstrates that their method is more effective at classifying cotton seed grades.

Figure 2. illustrates the image acquisition of cotton seeds and the extraction efficiency of the various features extracted from the images of the cotton seeds.

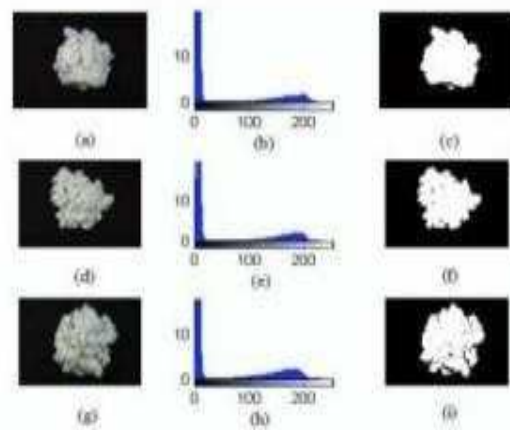


Fig. 2. Seed cotton image acquisition and extraction efficiency

In the reference [8], the authors presented a very innovative technique for segregation with the help of image processing. The innovative technique utilized various qualities to perform the segregation. The authors transmuted various qualities and features with the help of trichromatic shade space and spatial dependence grid. The authors attained a remarkable precision with the help of these innovative techniques and demonstrated that their method can effectively classify corn seed types.

Figure 3. includes the following:

- A camera to capture the images of the corn seeds.
- A light source to provide illumination for capturing the images of the corn seeds.
- A transparent sheet plate for placing the seed samples.
- A background plate.

Figure 4. illustrates the spatial relationships of the pixels by offsets and the pixel of interest for performing calculations

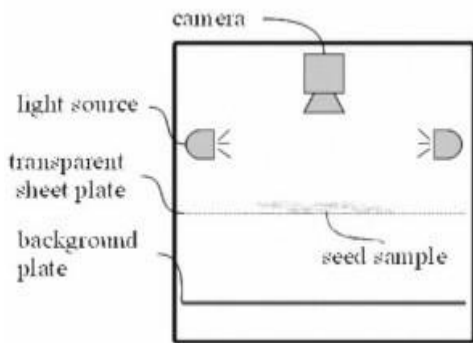


Fig. 3. Image capturing machine

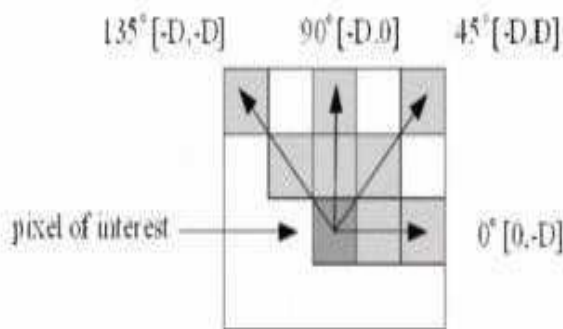


Fig. 4. Spatial relationships of pixel by offsets

In the reference [9], the authors propose a new method for rice variety classification using RGB and hyperspectral images. They demonstrate that this approach is more accurate than existing methods and could be used to improve rice breeding and other related applications. The authors demonstrated that their innovative technique can be of paramount importance in segregation.

Figure 5. displays a schematic diagram of the Hyperspectral Imaging (HSI) and Charge-Coupled Device (CCD) cameras being setup for performing data acquisition.

Figure 6. illustrates a block diagram for data processing task

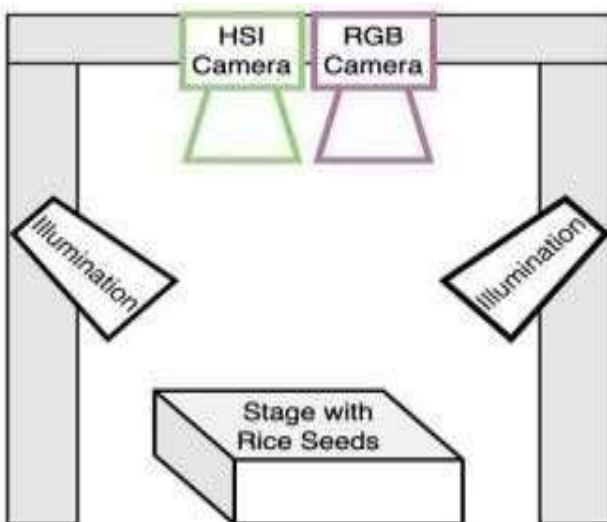


Fig. 5. A schematic diagram of the HSI and CCD cameras setup for data acquisition

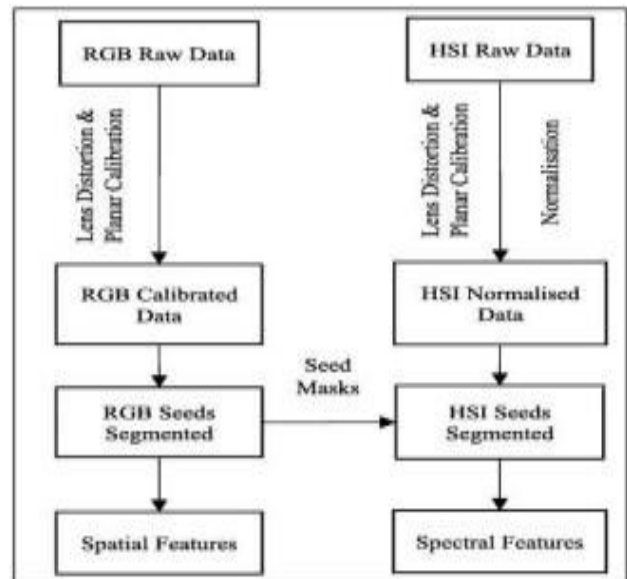


Fig. 6. A block diagram showing the stages of the data processing task

In reference [10], the authors propose a new machine learning approach for lettuce seed variety classification using single-kernel RGB images. They use support vector machines (SVMs) to learn the spectral features that distinguish different seed varieties. This approach is different from traditional image processing methods, which focus on extracting features from images using hand-crafted algorithms. The authors argue that their machine learning approach is more accurate and efficient, as it learns the features automatically from the data. The authors evaluate their approach on a dataset of lettuce seed images and achieve an accuracy of 97.5 %. This is significantly higher than the accuracy of 85 % achieved by existing image processing methods.

In the reference [11], the authors proposed a system for evaluating the germination rate of sorghum and lettuce seeds using lighting and photography. The system detects roots to estimate the germination rate but requires the roots to grow in a straight line. The system works by taking pictures of the seeds as they germinate. It then uses machine learning to identify the roots in the pictures. It then estimates the sprouting speed by performing numerous precise computations. The machine learning algorithm that the system uses to identify the roots is trained on images of roots that are growing in a straight line. If the roots do not grow in a straight line, the algorithm may not be able to identify them correctly.

In the reference [12], the authors presented an innovative technique that can correctly predict whether or not a rice seed will germinate with an accuracy of 92.34 %. However, the technique does not naturally approximate the sprouting speed. Moreover, the system can offer extra details about the requirements of grains. This innovative technique is exploited to predict whether the pips sprout or not.

Figure 7. illustrates the conceptual diagram of the Rice Seed Germination Evaluation System (RSGES).

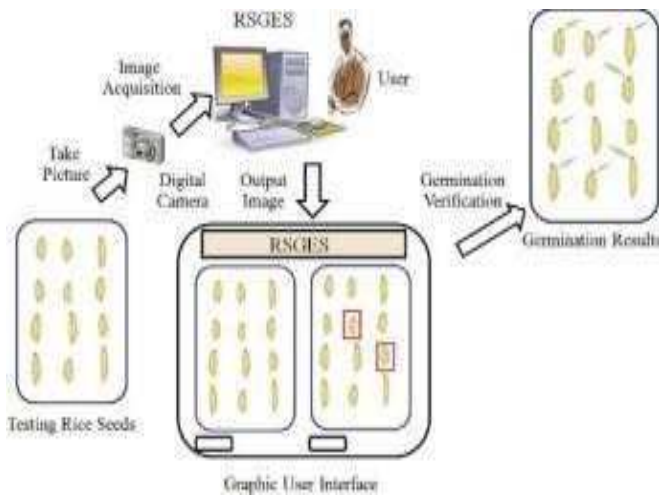


Fig. 7. Rice Seed Germination Evaluation System (RSGES) conceptual diagram

In reference [13], the authors propose a method for seed germination detection using HSV colour space and root detection. The proposed system first uses the HSV colour model to extract the root and seed regions of a plant. However, the system incorporates a technique which involves a very exhaustive and intensive pre-processing process. The system thrives to be extremely efficient with a remarkable precision rate.

In the reference [14], the authors examined an innovative forecasting technique which forecasts the sprouting rate of a glycine max pip throughout accommodation. They used a variable known as K1 which is the measure of the pip's resistance to deterioration. The authors further came to know that their innovative forecasting technique had better precision for pips which were recently accommodated. Finally, the authors resolved to the fact that the innovative forecasting technique is not a dependable technique and there is a need to create more efficient techniques.

In the reference [15], the authors developed a simulation model to predict how corn seeds germinate and emerge at different temperatures. The model accounts for water uptake, root and shoot elongation, and germination and emergence. The model was tested and found to be accurate at a range of temperatures in controlled environments and the field. However, it is not intended to predict responses to extreme temperatures (32°C) or to temperatures when other soil factors are limiting.

In the reference [16], the authors created a mathematical model to predict how many cucurbit seeds will germinate and how quickly they will germinate at different temperatures. The innovative technique takes into account the amount of days to germination, the temperature in degree celsius, and some coefficients that vary according to the type of cucurbit. The authors tested the model on six different types of cucurbits and found that it was able to accurately predict how many seeds would germinate and how quickly they would germinate at a range of temperatures. The model also predicted the best temperature for germination for each type of cucurbit. The best temperatures varied depending on the type of cucurbit but were all within the range of

temperatures that are typically used to germinate cucurbit seeds.

In the reference [17], the authors developed two new tests to predict how many radish seeds from a batch will produce normal seedlings. The first test, radicle emergence (RE) measures how quickly the root of the seed emerges. The second test, electrical conductivity (EC) measures the amount of electricity that can flow through the water in which the seeds have been soaked. The authors found that both the RE and EC tests were very accurate in predicting how many normal seedlings would be produced by a given batch of radish seeds. The tests were also quick and easy to perform. The authors conclude that the RE and EC tests could be used by radish seed producers and farmers to improve the germination rate and yield of radish crops.

In the reference [18], the authors created a dynamic artificial intelligence system to examine the sprouting of a pip. The system considered various qualities and features that are pivotal for examining the sprouting of a pip. The system demonstrated remarkable speed and precision which is reliable and highly efficient. The system incorporated rigorous practices and methodologies which helped it to attain high efficiency whilst being dependable.

In the reference [19], the authors created an efficient and innovative technique that takes into the cost of equipment, machinery, and tools. The authors then integrated a very efficient firmware to ensure the fast and efficient operation of the technique. The system is developed in such a way that it is capable of examining many pictures concurrently while also providing dependable information in the required structure. Finally, the authors were able to create a robust system that assures integrity and efficiency in the measurement of morphological features of several plant varieties which focuses on the study of physical form and structure.

In the reference [20], the authors delved into the details of the significant role played by different conditions in the sprouting estimation of pips. The authors tried to contrast the effects influenced by both stable conditions and varying conditions in the sprouting estimation of pips. Finally, the authors argued that there is a need for an extensive investigation as to whether the stable conditions contribute to poor sprouting estimation of pips.

III. INFERENCE FROM THE SURVEY

For the classification of seeds, many authors have used image processing algorithms like OpenCV and many Machine Learning and Deep Learning algorithms like K-means algorithm, CNN to avoid fraudulent transportation and sales in the rice industry. In classification, authors extract the features of seeds like textures, colour. The dataset varies from 350 to 10000. The best algorithm for classification is SVM (Support Vector Machine) with highest accuracy of 98.6 %. For germination, Machine Learning based on straight roots of seeds, HSV colour model, using mathematical model formula with required values and importantly image processing and computer vision using artificial neural network techniques were preferred. We inferred that these are not user friendly where users can't deal with these systems.

IV. CONCLUSION

This survey extensively covered the diverse array of approaches explored in this domain and articulated the persisting challenges that demand attention. Significantly, it underscored the immense potential that machine learning holds for precise grain type prediction. Furthermore, the review concluded by proposing future research directions, indicating a promising avenue for conducting in-depth data analysis on the dataset. This envisages simplifying data comprehension through techniques like Principal Component Analysis (PCA), followed by streamlining data further with K-Means clustering. Ultimately, the review highlights the importance of assessing accuracy, precision, and F1-Score metrics concerning the dataset, which could potentially provide valuable insights for scientists and researchers in this field. In conclusion, this literature review has provided valuable insights into the various factors influencing seed germination prediction. From examining the methodologies and models employed to explore the impact of environmental variables and genetic factors, it is evident that seed germination prediction is a complex and multifaceted field. Researchers have made significant strides in developing predictive tools and models, yet challenges remain in achieving precise and universally applicable predictions.

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